

Single URL Webscraper

This project is a command-line Python web scraper designed to extract meaningful content from a single URL. It uses the requests library to fetch web pages and BeautifulSoup to parse HTML and extract specific elements. Initially built to collect all hyperlinks from a page, the script was later enhanced to retrieve structured data such as quotes, authors, and tags from quotes.toscrape.com. The tool demonstrates the practical use of HTTP requests, HTML parsing, and command-line argument handling, and serves as a foundational exercise in automating data collection from static websites.

```
scrape_link.py > ...
1  import sys
2  import requests
3  from bs4 import BeautifulSoup
4
5  def scrape_links(url):
6      try:
7          response = requests.get(url)
8          response.raise_for_status()
9          soup = BeautifulSoup(response.text, 'html.parser')
10
11         links = soup.find_all('a', href=True)
12         if links:
13             print(f"\nLinks found on {url}:\n")
14             for link in links:
15                 print(link['href'])
16         else:
17             print(f"\nNo links found on {url}.")
18     except requests.exceptions.RequestException as e:
19         print(f"\nError fetching {url}:\n{e}")
20
21 if __name__ == "__main__":
22     if len(sys.argv) != 2:
23         print("Usage: python scrape_links.py <URL>")
24     else:
25         scrape_links(sys.argv[1])
26
```

My initial Python script takes a URL as a command-line argument, fetches the HTML content using requests, and parses it with BeautifulSoup to extract all hyperlink references (<a href> tags). It prints the list of links found on the page or displays an error message if the request fails or no links are present.

```
PS C:\single-url-webscrapper> python scrape_link.py https://quotes.toscrape.com/
Links found on https://quotes.toscrape.com/:
/
/login
/author/Albert-Einstein
/tag/change/page/1/
/tag/deep-thoughts/page/1/
/tag/thinking/page/1/
/tag/world/page/1/
/author/J-K-Rowling
/tag/abilities/page/1/
/tag/choices/page/1/
/author/Albert-Einstein
/tag/inspirational/page/1/
/tag/life/page/1/
/tag/live/page/1/
/tag/miracle/page/1/
/tag/miracles/page/1/
/author/Jane-Austen
/tag/aliteracy/page/1/
/tag/books/page/1/
/tag/classic/page/1/
/tag/humor/page/1/
/author/Marilyn-Monroe
/tag/be-yourself/page/1/
/tag/inspirational/page/1/
/author/Albert-Einstein
/tag/adulthood/page/1/
```

The script successfully ran from the command line and extracted all <a> tag links from the page at https://quotes.toscrape.com, displaying relative paths like author pages and tag filters. This confirms that the link-scraping functionality is working as intended.

```

✓ import sys
  import requests
  from bs4 import BeautifulSoup

✓ def scrape_quotes(url):
✓     try:
        response = requests.get(url)
        response.raise_for_status()
        soup = BeautifulSoup(response.text, 'html.parser')

        quotes = soup.find_all('div', class_='quote')

        for quote in quotes:
            text = quote.find('span', class_='text').get_text()
            author = quote.find('small', class_='author').get_text()
            tags = [tag.get_text() for tag in quote.find_all('a', class_='tag')]

            print(f'Quote: {text}')
            print(f'Author: {author}')
            print(f'Tags: {", ".join(tags)}\n')

        except requests.exceptions.RequestException as e:
            print(f"Error fetching {url}: {e}")

✓ if __name__ == "__main__":
✓     if len(sys.argv) != 2:
        print("Usage: python scrape_quotes.py <URL>")
✓     else:
        scrape_quotes(sys.argv[1])

```

The updated script extracts full quote data from the page, including the quote text, author name, and associated tags, instead of just links. It uses BeautifulSoup to target HTML elements with specific classes like quote, text, author, and tag to display clean, structured output.

```
PS C:\single-url-webscrapper> python scrape_link.py https://quotes.toscrape.com/
Quote: "The world as we have created it is a process of our thinking. It cannot be changed without changing our thinking
."
Author: Albert Einstein
Tags: change, deep-thoughts, thinking, world

Quote: "It is our choices, Harry, that show what we truly are, far more than our abilities."
Author: J.K. Rowling
Tags: abilities, choices

Quote: "There are only two ways to live your life. One is as though nothing is a miracle. The other is as though everyth
ing is a miracle."
Author: Albert Einstein
Tags: inspirational, life, live, miracle, miracles

Quote: "The person, be it gentleman or lady, who has not pleasure in a good novel, must be intolerably stupid."
Author: Jane Austen
Tags: aliteracy, books, classic, humor

Quote: "Imperfection is beauty, madness is genius and it's better to be absolutely ridiculous than absolutely boring."
Author: Marilyn Monroe
Tags: be-yourself, inspirational

Quote: "Try not to become a man of success. Rather become a man of value."
Author: Albert Einstein
Tags: adulthood, success, value
```

The script successfully scraped and printed each quote, its author, and associated tags from quotes.toscrape.com, confirming accurate extraction of structured content. This output demonstrates the scraper is now parsing the core text content of the site, not just the links.

Quotes to Scrape

"The world as we have created it is a process of our thinking. It cannot be changed without changing our thinking."

by [Albert Einstein](#) ([about](#))

Tags: [change](#) [deep-thoughts](#) [thinking](#) [world](#)

"It is our choices, Harry, that show what we truly are, far more than our abilities."

by [J.K. Rowling](#) ([about](#))

Tags: [abilities](#) [choices](#)

"There are only two ways to live your life. One is as though nothing is a miracle. The other is as though everything is a miracle."

by [Albert Einstein](#) ([about](#))

Tags: [inspirational](#) [life](#) [live](#) [miracle](#) [miracles](#)

"The person, be it gentleman or lady, who has not pleasure in a good novel, must be intolerably stupid."

by [Jane Austen](#) ([about](#))

Tags: [aliteracy](#) [books](#) [classic](#) [humor](#)

"Imperfection is beauty, madness is genius and it's better to be absolutely ridiculous than absolutely boring."

by [Marilyn Monroe](#) ([about](#))

Tags: [be-yourself](#) [inspirational](#)

"Try not to become a man of success. Rather become a man of value."

by [Albert Einstein](#) ([about](#))

Tags: [adulthood](#) [success](#) [value](#)

"It is better to be hated for what you are than to be loved for what you are not."

by [André Gide](#) ([about](#))

This is the site being scraped for reference. Quotes To Scrape is a website designed to test web scrappers on, it is important to note that many sites do not allow webscrapping in their terms of service.